

Communication System

0 Review Of Signal & System

0.1 Definition and types: basic and composite signals, dimensions: single, multi Channel: single, multi

0.2 Analog and Digital signals: Continuous and Discrete time signals

0.2.1 Classification of Signals: i) Periodic and Non-periodic Signals ii) Even and Odd signals iii) Deterministic and Non-Deterministic Signals iv) Energy and Power Signals, Continuous Time and Discrete Time Signals v) Causal, Non-causal and Anti-causal signal

0.3 Systems

0.3.1 Basic Properties

0.3.2 Classifications: i) Static and Dynamic ii) Linear and non-linear iii) Time variant and invariant iv) Causal and non-causal v) Stable and unstable vi) Feedback and non-feedback

0.4 Fourier series and Fourier Transform

0.4.1 Conditions, properties, Frequency domain representation (magnitude and phase plot)

0.5 Convolution & Correlation

1 Introduction

1.1 Communication System

1.2 Block diagram of analog and digital communication systems in Different Transmission Modes (simplex , duplex & half duplex)

1.3 Transmission/Communication Impairments : Noise, attenuation, and interference

1.4 Performance of network (Bandwidth, throughput, latency, jitter)

1.5 Coding : Channel And Source

1.6 Transmission modes: Simplex, half-duplex, full-duplex

1.7 Conversation : Analog-Analog, Analog-Digital, Digital-to-Analog, and Digital-to-Digital signal encoding Techniques

1.8 Compression Of Data : Lossless (Hauffman & Shannon Fano) & Lossy

1.9 Modulation : Types And Need

1.10 Channel capacity : Nyquist bandwidth, Shannon-Hartley capacity Theorem

2 Amplitude Modulation

2.1 Time domain expressions, frequency domain representation, modulation index, signal bandwidth of amplitude modulated signal

2.2 AM for single and double tone message, carrier and sideband components, power in carrier and sideband components, bandwidth and power efficiency, Hilbert transform

2.3 Double sideband AM (DSB-FC), generation (Square law), detection (Envelope and square law method)

- 2.4 Double sideband suppress carrier (DSB-SC), Generation (Linear modulator, balance modulator), Synchronous detection method
- 2.5 Overview of SSB, VSB, and ISB modulations
- 2.6 Phase locked loop (PLL), demodulation of AM using PLL
- 2.7 Super-heterodyne AM receiver

3 Angle Modulation

- 3.1 Basic definition, time domain expression for frequency modulation (FM) and phase modulation (PM)
- 3.2 Time domain expression for single tone, modulated FM signals, spectral representation
- 3.3 Bandwidth of FM, Carson's rule, narrow and wideband FM
- 3.4 Generation of FM: Direct and Armstrong's methods
- 3.5 Demodulation of FM signals: Non-synchronous (Limiter discriminator) and synchronous (PLL)
- 3.6 Stereo FM, spectral details, pre-emphasis and de-emphasis network
- 3.7 Super-heterodyne Radio receiver for FM

4 Pulse Modulation

- 4.1 Fundamental of Analog Pulse Modulation : PAM , PWM , PPM
- 4.2 Digital Pulse Modulation : PCM , DPCM , DM
- 4.3 Pulse coded modulation (PCM)
 - 4.3.1 Sampling (Sampling theorem, ideal sampling, practical sampling, aliasing effect, aperture effect, signal reconstruction)
 - 4.3.2 Quantization
 - 4.3.2.1 Uniform Quantization - Mid riser & Mid Trend
 - 4.3.2.2 Non-uniform quantization (Companding techniques) : A-law & μ -Law
 - 4.3.3 Encoding
- 4.4 DPCM : Encoder , Decoder , comparison with PCM
- 4.5 DM: Encoder, decoder, advantage, disadvantage, noise in DM (Granular noise & Slope Overloading Error)

5 Channel Bandwidth Utilisation

- 5.1 Multiplexing
 - 1.1 TDM
 - 1.2 FDM
 - 1.3 WDM

5.2 Switching

- 2.1 Circuit Switching
- 2.2 Message Switching
- 2.3 Packet Switching

5.3 Multiple Access

- 3.1 TDMA : Synchronous TDMA and Statistical TDMA
- 3.2 FDMA
- 3.3 SDMA
- 3.4 CDMA : Near-far problem in CDMA & Diagram of 4 stations communicating with each other using CDMA

5.4 Hierarchies

- 4.1 FDM Analog
- 4.2 TDM Digital
 - 4.2.1 T1 (North American Digital T-carrier) Hierarchy
 - 4.2.2 E1 (European digital multiplexing) Hierarchy

6 Baseband Digital Data Transmission

- 6.1 Information theory, measurement of information, entropy, symbol rates and data rates
- 6.2 Shannon Hartley channel capacity theorem, implication of theorem, and theoretical limits
- 6.3 Compression techniques: Shannon-Fano, Huffman codes
- 6.4 Line coding schemes:
 - 6.4.1 Unipolar
 - 6.4.2 Polar
 - 6.4.2.1 NRZ : NRZ-I , NRZ-L
 - 6.4.2.2 RZ
 - 6.4.2.3 Biphase : Manchester, Differential Manchester,
 - 6.4.3 Bipolar : AMI
 - 6.4.4 Scrambling techniques: B8ZS, HDB3
- 6.5 ISI, Nyquist criteria, pulse shaping for zero ISI

7 Bandpass Digital Modulation Techniques

- 7.1 ASK, FSK, PSK :generation, properties, constellation diagram and detections
- 7.2 Constellation Diagram & Eye Diagram
- 7.3 M-PSK : Generation , Constellation Diagram
- 7.4 M-QAM : Generation , Constellation Diagram
- 7.5 Minimum Shift Keying (MSK)
- 7.6 Gaussian Minimum Shift Keying (MSK)

7.7 Constellation diagram of ASK, FSK, PSK, QPSK, 8-QAM, 16-QAM, 64-QAM

8 Error CONTROL Coding

8.1 Basic Terms Used In Error Control : Hamming weight, hamming distance, code vectors, constraint length, code rate, syndromes

8.2 Error detection

8.2.1 Parity Check

8.2.2 Cyclic Codes

8.2.3 Checksum

8.2.4 CRC-n ($n = 3$ to 32)

8.3 Error correction codes:

8.3.1 Linear block codes

8.3.2 Hamming codes

8.3.3 Convolutional code

1. Encoding

1.1 State Space Diagram

1.2 Code Tree Diagram

1.3 Trellis Diagram

2. Decoding

2.1 Sequential Decoding

2.2 Viterbi Decoding / Maximum Likely-hood Decoding

9 Noise in Communication Systems

9.1 Definition, white noise, AWGN channel, PSDF, and AC function of white noise

9.2 Ideal low-pass and RC filtering of white noise, noise equivalent bandwidth of a filter

9.3 Optimum detection of a pulse in additive white noise, the matched filter, realization of matched filters (Time correlators), the matched filter for a rectangular pulse

9.4 Overview of error probability function in digital communication (ASK, FSK, and PSK)